



Original Article

Sleep disorders and daytime sleepiness in children with attention-deficit/hyperactivity disorder: A two-night polysomnographic study with a multiple sleep latency test

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ABSTRACT

Objective: To evaluate sleep macrostructure, sleep disorders incidence and daytime sleepiness in attention-deficit/hyperactivity disorder (ADHD) affected children compared with controls.

Methods: Thirty-one patients (26 boys, 5 girls, mean age 9.3 ± 1.7 , age range 6–12 years) with ADHD diagnosed according to DSM-IV criteria, without comorbid psychiatric or other disorders, as never before pharmacologically treated for ADHD. The controls were 26 age- and sex-matched children (22 boys, 4 girls, age range 6–12 years, mean age 9.2 ± 1.5). Nocturnal polysomnography (PSG) was performed for two nights followed by the multiple sleep latency test (MSLT).

Results: No differences between the two groups comparing both nights were found in the basic sleep macrostructure parameters or in the time (duration) of sleep onset. A first-night effect on sleep variables was apparent in the ADHD group. Occurrence of sleep disorders (sleep-disordered breathing [SDB], periodic limb movements in sleep [PLMS], parasomnias) did not show any significant differences between the investigated groups. A statistically significant difference ($p = 0.015$) was found in the trend of the periodic limb movement index (PLMI) between two nights (a decrease of PLMI in the ADHD group and an increase of PLMI in the control group during the second night). While the mean sleep latency in the MSLT was comparable in both groups, children with ADHD showed significant (sleep latency) inter-test differences (between tests 1 and 2, 1 and 4, 1 and 5, $p < 0.01$).

Conclusion: After the inclusion of adaptation night and exclusion of psychiatric comorbidities, PSG showed no changes in basic sleep parameters or sleep timing, or in the frequency of sleep disorders (SDB, PLMS) in children with ADHD compared with controls, thus not supporting the hypothesis that specific changes in the sleep macrostructure and sleep disturbances are connected with ADHD. A first-night effect on sleep variables was apparent only in the ADHD group. Though we found no proof of increased daytime sleepiness in children with ADHD against the controls, we did find significant vigilance variability during MSLT in the ADHD group, possibly a sign of dysregulated arousal.

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1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) is the most frequent neurobehavioral disorder of childhood with an estimated prevalence of 3–5% of all school children [1]. It is characterised by developmentally inappropriate symptoms of inattention, hyperactivity and impulsivity that begin in early childhood. According to the American Psychiatric Association diagnostic guideline (Diagnostic and Statistical Manual, version IV [DSM-IV]) [2], ADHD is

classified into three subtypes: predominantly inattentive, predominantly hyperactive-impulsive and combined. The diagnosis of ADHD is based on clinical evaluation of a child's behavior and on the criteria of DSM-IV [2].

According to questionnaire studies, sleep problems are so common in children with ADHD (in 50–60%) [3–5] that they were also included in the diagnostic criteria of the previous Diagnostic and Statistical Manual of Mental Disorders (DSM-III) [6]. This fact inspired the hypothesis that sleep disorders in ADHD might be caused by the same neurochemical and structural abnormality responsible for the development of ADHD, i.e., a disorder of neurotransmitters (mainly dopamine and noradrenaline) as well as by structural abnormalities in the prefrontal cortex and those subcortical areas which are important for the control of behavioral

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responses such as attention and executive function but also for the control of arousal [7].

The results of objective studies for the detection of ADHD-specific sleep disorders are rather heterogeneous. Some authors accordingly found rapid eye movement (REM) sleep reduction and longer REM sleep latency in children with ADHD in what they believe is an ADHD-specific feature [8–12]. However, according to overviews and meta-analytic studies, no specific sleep macrostructure change has yet been proved [13–17], while changes in the duration and timing of sleep (as reported in questionnaire studies) appear to be influenced rather by emotional and behavioral factors [13,15]. To go by other studies, the specific intrinsic problem of underlying ADHD rests in arousal dysregulation [12,18]. This idea is supported in two studies which found evidence of hypoarousal state in increased daytime sleepiness in children with ADHD [19,20] and in another study which discovered abnormalities in the microstructure of sleep-reduced cyclic alternating patterns in patients with ADHD [21].

Currently under discussion is the relationship between ADHD and sleep disturbances, especially sleep-disordered breathing (SDB) and periodic limb movements in sleep/periodic limb movement disorder (PLMS/PLMD) as these disorders of sleep coupled with sleep disruption are known to cause behavioral and cognitive disorders that may mimic ADHD [16,22]. Present-day studies fall into those which find a higher rate of SDB [20,23–25] and PLMS/PLMD [20,25–31] in children with ADHD to speculate about their crucial significance in this disease, and those in which ADHD is seen as an independent entity different from ADHD-like symptoms caused by sleep disorders connected with sleep fragmentation [8,14–16,22,32–34]. The authors of latter studies presume that sleep disorders in most patients with ADHD have no major role to play in the pathogenesis of this disorder.

The outcome of PSG studies of ADHD depends on the choice of the cohort, on the presence of comorbidities, current medication with psychostimulants and other drugs, on methodological limitations such as the absence of the adaptation night, on insufficient testing for concomitant psychiatric disorders (particularly anxiety and depression) that can influence sleep, or on the absence of controls [13–16].

The aim of our study was to evaluate the sleep macrostructure including an adaptation night, but also the rate of sleep disorders and level of daytime wakefulness in the ADHD group, without concurrent psychiatric involvement and without pharmacological treatment. The data obtained were compared with controls.

2. Subjects and methods

2.1. Participants

Our study subjects were recruited from among outpatients of paediatric neurologists and psychiatrists and referred to our departments of psychiatry and neurology where the study was conducted. Recruitment was aimed at children with ADHD coming for initial psychiatric assessment before stimulant or other pharmacotherapy was started. In total, 31 children with diagnosed ADHD (26 boys, 5 girls, mean age 9.3 ± 1.7 , age range 6–12 years, body mass index-BMI 18.1 ± 2.5) were enrolled. The enrollment criteria were as follows: (1) ADHD diagnosed on the basis of DSM-IV [2], (2) no previous pharmacological treatment for ADHD, (3) no history of any chronic physical condition (including obesity), chronic sleep disorder, neurological or other psychiatric disorders (including mental retardation and autism) based on a complete pediatric report and on a neurological and psychiatric examination, (4) no current medication (psychotropic or general) and (5) the patient's and his/her parents' willingness to participate in the study and informed consent signed by the parents.

All participants were in the prepubertal or early pubertal stages as assessed by Tanner scale (Tanner stage 1–2) [35]. All were of Caucasian origin. They underwent psychological, psychiatric and neurological testing. Information about their sleep habits and sleep disturbances was collected from the parents and children by means of a detailed clinical interview and Pediatric Sleep Questionnaire [36]. Reported sleep problems were not taken as a reason for exclusion.

The diagnosis of ADHD was established by means of a detailed clinical interview which included a structured psychiatric examination (Children's Psychiatric Rating Scale) [37] and DSM-IV diagnostic criteria for ADHD [2]. These criteria were also used for the classification of ADHD subtypes. A combined-type of ADHD was predominant (27 patients), and four had inattentive type of ADHD. Parentally reported onset of symptoms was between 4 and 6 years in all children. The Conners' Parent Rating Scale [38] was also employed to assess the severity of ADHD symptoms (a score of at least 2 SD above the mean on this scale and ADHD index were used to classify the children as having significant ADHD symptoms). To exclude emotional and behavioral problems the following additional tools were employed: Child Behavior Checklist for parents, Children's Manifest Anxiety Scale and Children's Depression Inventory. The IQ level was assessed by means of the Wechsler Intelligence Scale for Children-Revised (WISC-III); all children had $IQ > 80$, and WISC-III full scale values were 104.9 ± 12.3 .

The control group was recruited from the same district area (through schoolmates of participants with ADHD, children of friends and relatives of medical staff) and was made up of 26 healthy non-obese prepubertal or early pubertal (Tanner stage 1–2) children matched for age and sex, 22 boys and 4 girls, age range 6–12 years, mean age 9.2 ± 1.5 , BMI 17.5 ± 1.4 . All were without any chronic disease (as established from a complete pediatric report) and without any medication. The diagnosis of ADHD was ruled out according to DSM-IV criteria. The Conners' Parent Rating Scale and Child Behavior Checklist were also administered. The IQ level was not examined in the control group (all children in the control group had moderate or above moderate school achievements). Information on the subjects' sleep habits and disorders was obtained from the Pediatric Sleep Questionnaire and in detailed clinical interviews with the parents and children. The results of psychiatric scales are summarized in Table 1.

2.2. Measures

Children with ADHD and the controls had nocturnal video-polysomnography (PSG) performed for two consecutive nights in the sleep laboratory followed by a multiple sleep latency test (MSLT). The first one was rated as an adaptation night. The children's bedtime was very much like their usual bedtime back home as was their usual weekday regimen rising time after each PSG, i.e., between 6:30 and 7 a.m.

A Schwarz polygraph was used for all polysomnography studies with standard electroencephalographic montage (F4–C4, C4–P4, F3–C3, C3–P3, C4–A1 and C3–A2), horizontal electrooculography, submental and bilateral anterior tibialis electromyography, electrocardiography and video recording using an infrared-light

Table 1
Characteristics of participants: results of psychiatric scales.

Psychiatric scale	ADHD ($n = 31$)	Controls ($n = 26$)	P value t -test
CPRS global index total	76.9 ± 13.1	46.7 ± 5.3	$p < 0.001$
ADHD index	86.5 ± 17	46.4 ± 2	$p < 0.001$
CBCL total	73.8 ± 17.3	47.5 ± 8.1	$p < 0.001$

T -scores are provided. Data are presented as mean $\pm$ standard deviation.

camera. Oronasal airflow was monitored with thermistors. Thoracic and abdominal respiratory movements were recorded using belts with piezo sensors. Respiratory sounds were microphone-monitored. Oxyhemoglobin saturation was measured using pulse oximetry. Neither esophageal pressure measurement nor nasal cannula/pressure transducer was used. The employment of nasal cannula/pressure transducer offers a more accurate method for the evaluation of subtle forms of SDB, i.e., for the detection of upper airway resistance syndrome.

Taking into account possible night-to-night variability of sleep pattern in children with ADHD [12] and potential first-night effect on sleep architecture, a sleep analysis of the adaptation night was also obtained. Sleep stages were visually scored according to the Rechtschaffen and Kales standard criteria in 30-s epochs [39]. Conventional sleep parameters were evaluated (total sleep time, sleep efficiency, sleep-onset latency, REM sleep latency, percentage of sleep stages and movement time). Apneas and hypopneas were scored following definitions to determine the apnea index (AI) and apnea-hypopnea index (AHI) in children [40]. AHI > 1 was considered abnormal on the basis of normative data [41]. This more conservative criterion for SDB assessment was chosen mainly with regard to reported connection between mild forms of SDB and symptoms of ADHD [8,22,23,42]. Periodic limb movements were scored according to standard ICSD-2 criteria [43], except for the duration, which we defined as 0.5–10 s following recommendations from American Academy of Sleep Medicine and the International Restless Legs Syndrome Study Group [44]. The periodic limb movement index (PLMI) was calculated as the number of PLM per hour of sleep. As a cut-off for abnormality we chose PLMI > 5 per hour of sleep, a value generally considered abnormal in children [43]. PLMS associated with breathing events were not scored; only those independent of apnea/hypopnea were considered. PLMS were evaluated during both nights with respect to their potential night-to-night variability. The other sleep disorders, i.e., SDB and parasomnia (disorders of arousal from NREM sleep), were assessed only in the second night.

The multiple sleep latency test was performed during the day following the second PSG in all children with ADHD and in 25 controls (1 of the controls missed the test). Standard methods were used for MLST evaluation where the occurrence of one epoch of sleep of any stage was rated as sleep onset [45]. The mean sleep-onset latency was calculated. MLST recording times were at 10 a.m., 12 a.m., 2 p.m., 4 p.m. and 6 p.m., each test lasting 20 min. We chose 20-min MLST tests because that duration was used in most of the studies of daytime sleepiness in children [46–49].

2.3. Data analysis

Basic statistical characteristics—means and standard deviations—were calculated for continuous variables. As for discrete variables, the share of each of the categories was expressed in percent. To test the difference in the mean values between the ADHD and controls, the two-sample *t*-test was employed. The chi-square test or Fisher's exact test in frequency tables were used for comparisons of the relative frequencies. MSLT measurements were assessed by ANOVA (analysis of variance) with repeated measures and grouping factor because of the possibility to analyze in one model simultaneously the effects of the grouping factor, i.e., ADHD and repetition during MSLT as well as interaction between these two factors. Microsoft Excel, BMDP Statistical Software, MedCalc and STATISTICA 8.0 were used for statistical analysis. Pearson's correlation analysis was used for comparison between the polysomnographic/MSLT parameters and the data from psychiatric scales (Conners' Parent Rating Scale, Child Behavior Checklist).

3. Results

3.1. Polysomnographic findings

The basic sleep macrostructure parameters revealed no differences between the groups either the first or the second night (Table 2). The same applied to bedtime/sleep onset.

Comparing the first and the second nights for sleep variables within each group, significantly different trends were apparent between the ADHD and the control group in the following parameters: wakefulness ($p = 0.04$), sleep efficiency ($p = 0.03$), sleep latency ($p = 0.03$) and PLMI ($p = 0.015$) by means of the first step in repeated measures ANOVA. In the ADHD group a significant increase in wakefulness ($p = 0.0002$), a reduced sleep efficiency ($p = 0.00016$) and a prolonged sleep-onset latency ($p = 0.002$) were observed during the first night when compared with the second night. In the control group no significant inter-night changes were apparent.

The occurrence of sleep disorders (SDB, parasomnias) rated during the second night revealed no significant inter-group differences (Table 3). The AI and AHI were similar in both groups. Though PLMS did show discernible inter-night variability of occurrence, the differences never reached any statistical significance, nor did the PLMI values. The significant difference was found in the trend of PLMI between the two nights (a PLMI decrease in the ADHD group and a PLMI rise in the control group during the second night).

Nineteen children with ADHD and 15 controls had no PSG evidence of either SDB or PLMS, and comparisons of the sleep macrostructure parameters between these groups did not reveal any significant differences.

3.2. MSLT results

The MSLT-established mean sleep latency showed no inter-group differences on comparing the whole ADHD and control group (Table 4). Comparisons between subgroups without SDB/PLMS revealed no such differences either, and the two groups (ADHD and controls) showed no difference in the number of children who fell asleep while being tested (Table 4). REM sleep was noted in only one test in either group.

As for the ADHD group, a significant inter-test variability was observed. Sleep latency values displayed statistically significant differences between some of the tests during the day such as between tests 1 and 2, 1 and 4 and 1 and 5 ($p < 0.01$). Borderline changes were noted between tests 3 and 4 and 3 and 5 ($p = 0.061$) (Graph 1). Similar changes were also apparent in children with ADHD and normal PSG: a significant change between tests 1 and 2, 1 and 4 ($p < 0.01$), between tests 1 and 5, 3 and 5 ($p < 0.05$), and a borderline change between tests 3 and 4 ($p < 0.07$) (Graph 2). In other words, sleep latency during MSLT in children with ADHD exhibited significant time-related changes compared with the control group. No such changes were noted in the controls.

3.3. Comparison between polysomnographic variables, MSLT results and psychiatric scales

Correlation analysis of the study cohort revealed no statistically significant interdependence between the sleep onset latencies during MSLT and the scores of the psychiatric scales used (Conners' Parent Rating Scale and Child Behavior Checklist), and there was no significant correlation found between the scores of these psychiatric scales and the sleep variables.

3.4. Subjective sleep assessment

Subjective data on sleep habits and disturbances (as extracted from detailed clinical interviews and Pediatric Sleep Question-

Table 2
Polysomnographic characteristics.

Sleep variables	First night		<i>P</i> value <i>t</i> -test	Second night		<i>P</i> value <i>t</i> -test
	ADHD (<i>n</i> = 31)	Controls (<i>n</i> = 26)		ADHD (<i>n</i> = 31)	Controls (<i>n</i> = 26)	
*Sleep latency (min)	23.2 ± 18.6	16.4 ± 13.9	<i>p</i> = 0.512	13.0 ± 11.0	14.9 ± 10.6	<i>p</i> = 0.512
Sleep onset time (hours:min)	21:56	22:09	N/A	21:55	21:40	N/A
Sleep cycles (no)	5.2 ± 0.9	5.2 ± 0.6	<i>p</i> = 0.25	5.3 ± 0.6	5.5 ± 0.7	<i>p</i> = 0.25
REM latency (min)	119.8 ± 49.4	111.0 ± 43.1	<i>p</i> = 0.98	102.3 ± 36.9	102.5 ± 42.6	<i>p</i> = 0.98
Total sleep time (min)	505.5 ± 56.2	481.9 ± 37.4	<i>p</i> = 0.61	520.5 ± 33.7	524.4 ± 29.5	<i>p</i> = 0.61
*Sleep efficiency (%)	89.2 ± 6.4	91.2 ± 5.1	<i>p</i> = 0.42	94.6 ± 3.5	93.8 ± 3.9	<i>p</i> = 0.42
*Wakefulness (%)	6.6 ± 5.7	4.3 ± 4.4	<i>p</i> = 0.42	2.4 ± 1.8	2.9 ± 2.8	<i>p</i> = 0.42
NREM 1 (%)	5.1 ± 4.7	3.1 ± 2.3	<i>p</i> = 0.056	2.9 ± 3.0	1.7 ± 1.1	<i>p</i> = 0.056
NREM 2 (%)	40.8 ± 8.9	43.5 ± 6.7	<i>p</i> = 0.56	43.1 ± 4.8	43.8 ± 4.2	<i>p</i> = 0.56
NREM 3, 4 (%)	26.9 ± 9.1	26.3 ± 7.0	<i>p</i> = 0.93	27.0 ± 4.0	27.1 ± 3.9	<i>p</i> = 0.93
REM (%)	20.5 ± 4.6	21.6 ± 3.7	<i>p</i> = 0.83	24.0 ± 3.8	23.8 ± 3.3	<i>p</i> = 0.83
Movement time (%)	0.1 ± 0.2	0.2 ± 0.3	<i>p</i> = 0.50	0.4 ± 0.7	0.3 ± 0.3	<i>p</i> = 0.50
AHI	1.8 ± 3.2	1.3 ± 1.2	<i>p</i> = 0.83	1.4 ± 3.1	1.0 ± 1.1	<i>p</i> = 0.83
AI	1.1 ± 1.5	0.5 ± 0.8	<i>p</i> = 0.88	0.6 ± 1.0	0.6 ± 0.7	<i>p</i> = 0.88
PLMI	4.2 ± 4.3	2.8 ± 3.5	<i>p</i> = 0.26	2.9 ± 4.6	4.8 ± 7.8	<i>p</i> = 0.26

Data are presented as mean ± standard deviation.

*Significantly different trends were apparent between the first and second night in the ADHD group: wakefulness (*p* = 0.04), sleep efficiency (*p* = 0.03), sleep latency (*p* = 0.03) and PLMI (*p* = 0.015) by means of ANOVA with repeated measures.

Table 3
Occurrence of sleep disorders.

Sleep disorders	ADHD (<i>n</i> = 31)	Controls (<i>n</i> = 26)	<i>P</i> value Fisher's test
Sleep-disordered breathing (AHI > 1)	10 (32.2)	5 (19.2)	<i>p</i> = 0.37
Periodic limb movements in sleep (PLMI > 5) 1st night	9 (29.0)	4 (15.3)	<i>p</i> = 0.34
Periodic limb movements in sleep (PLMI > 5) 2nd night	5 (16.1)	8 (30.7)	<i>p</i> = 0.22
Parasomnia	8 (25.8)	3 (11.5)	<i>p</i> = 0.20
Bruxisms	2 (6.4)	2 (7.6)	<i>p</i> = 1.00

Data are presented as number (percentage).

Table 4
Results of multiple sleep latency test.

MSLT	Sleep onset latencies*			Children who fell asleep**		
	ADHD (<i>n</i> = 31)	Controls (<i>n</i> = 25)	<i>P</i> value <i>t</i> -test	ADHD (<i>n</i> = 31)	Controls (<i>n</i> = 25)	<i>P</i> value Fisher's test
Mean sleep latency (min)	16.1 ± 3.4	17.1 ± 2.4	<i>p</i> = 0.23	–	–	–
1st test	13.2 ± 6.6	16.0 ± 5.3	<i>p</i> = 0.087	20 (64.5)	11 (42.3)	<i>p</i> = 0.18
2nd test	16.9 ± 5.1	17.4 ± 4.3	<i>p</i> = 0.69	11 (35.4)	8 (30.7)	<i>p</i> = 1.00
3rd test	15.3 ± 5.8	17.2 ± 4.5	<i>p</i> = 0.18	14 (45.1)	8 (30.7)	<i>p</i> = 0.41
4th test	17.3 ± 4.5	17.9 ± 3.5	<i>p</i> = 0.58	10 (32.2)	8 (30.7)	<i>p</i> = 1.00
5th test	17.9 ± 4.8	17.1 ± 2.4	<i>p</i> = 0.44	6 (20.0)	8 (30.7)	<i>p</i> = 0.36

* Data are presented as mean ± standard deviation.

** Data are presented as number (percentage).

naire) provided information mainly on restless sleep (*p* = 0.001), sleeplessness with sleep onset coming as late as 1–1.5 h (*p* = 0.047) and leg kicking in sleep (*p* = 0.013) in the ADHD group compared with controls (Table 5). These subjective complaints were inconsistent with the actual PSG findings. Although ten of the ADHD-affected children reported difficulty with and prolongation of sleep, none of them had a sleep-onset latency longer than 30 min, which rules out sleep onset insomnia. As for leg kicking, only two in the ADHD group with this problem had a PLMI > 5 on PSG.

4. Discussion

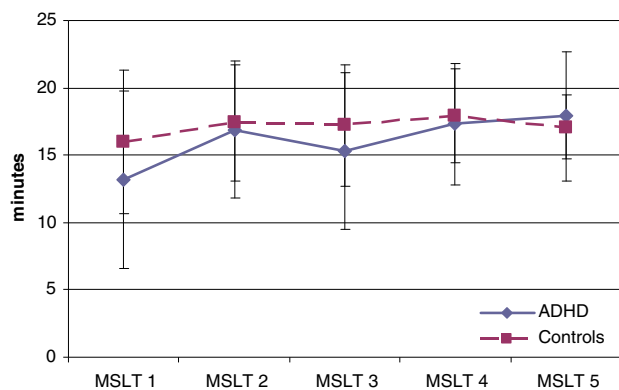
Our study was aimed at an objective evaluation of sleep macrostructure (during two nights of polysomnography), sleep disorders and daytime sleepiness in children with ADHD without psychiatric or other comorbidities and never before pharmacologically treated for ADHD.

Although subjective reports on sleep disturbances revealed in ADHD children significantly more complaints of restless sleep, dif-

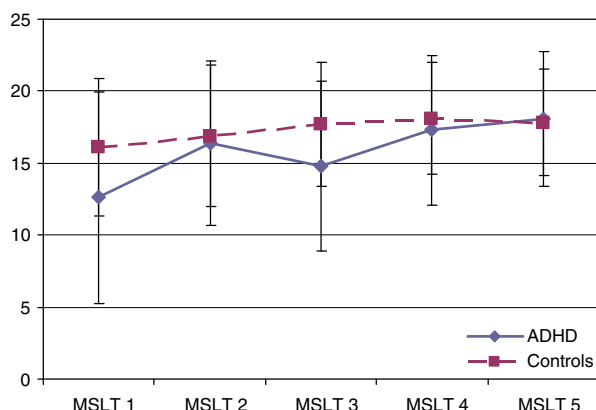
Table 5
Subjectively reported sleep problems.

Complaints	ADHD (<i>n</i> = 31)	Controls (<i>n</i> = 26)	<i>P</i> value Fisher's test
Restless sleep	15	2	<i>p</i> = 0.001
Insomnia (difficulty in falling asleep)	10	2	<i>p</i> = 0.047
Difficult morning awakening	9	3	<i>p</i> = 0.19
Nocturnal awakening	1	1	<i>p</i> = 1.00
Bedtime resistance	2	0	<i>p</i> = 0.49
Parasomnia (somnambulism, sleep terrors)	6	3	<i>p</i> = 0.49
Bruxisms	3	2	<i>p</i> = 1.00
Ronchopathy	11	7	<i>p</i> = 0.58
Sleep apnea	1	0	<i>p</i> = 1.00
Leg kicking	7	0	<i>p</i> = 0.013
Sleep talking	7	4	<i>p</i> = 0.74
Nightmares	2	3	<i>p</i> = 0.65
Dry mouth in the morning	2	0	<i>p</i> = 0.49
Restlessness before sleep, leg movements	2	0	<i>p</i> = 0.49
Growing pains	0	1	<i>p</i> = 1.00

Data are presented as number.



Graph 1. Results of multiple sleep latency test – intergroup comparison between the ADHD and control group. Significant sleep latency changes between tests 1 and 2, 1 and 4, 1 and 5 ($p < 0.01$) and borderline changes between tests 3 and 4, 3 and 5 ($p = 0.061$) were apparent in the ADHD group.



Graph 2. Results of multiple sleep latency test – intergroup comparison between the ADHD and control group with normal polysomnography. Significant sleep latency changes between tests 1 and 2, 1 and 4 ($p < 0.01$), between test 1 and 5, 3 and 5 ($p < 0.05$) and a borderline change between tests 3 and 4 ($p < 0.07$) were apparent in the ADHD group.

difficulties in initiating sleep, and leg kicking than in the controls, these reported data were not consistent with the actual PSG findings. The fact that subjective data on the quality and length of sleep and its disorders in children with ADHD are at variance with objectively established facts and figures was repeatedly observed [3,4,10]. This is logical because objective methods measure variables which are very different from those assessed with subjective ones.

The data from both PSG were analyzed in order to evaluate the first-night effect on sleep architecture in both groups and to assess night-to-night variability of the sleep pattern previously reported in children with ADHD [18,31,50,51]. PSG during both nights showed no changes in conventional sleep parameters while comparing the whole ADHD group under study with the controls or the subgroups without the occurrence of SDB/PLMS. There was no evidence of any significant changes in either bedtime or sleep onset such as tendency for sleep delay in children with ADHD. A first-night effect was apparent in the ADHD group as a significant increase in wakefulness, reduced sleep efficiency and prolonged sleep latency compared to the second night, but no changes of REM sleep or its latency were marked. Thus, we were unable to prove the hypothesis that REM sleep reduction such as some authors found in children with ADHD [8–12] might possibly be

attributable to first-night effect. But then, these REM sleep changes may have been influenced by psychiatric comorbidities [8–10,12] which we had excluded from our cohort.

The high degree of night-to-night variability in sleep-related behavior and sleep patterns in ADHD children observed in questionnaire-based and actigraphic studies [18,31,50,51] led Gruber et al. [12] to speculate an intrinsic circadian rhythm disorder there. To what extent intrinsic as distinct from behavioral and environmental factors are responsible for chronic sleep deprivation and irregular sleep pattern is a moot point. Although our evaluation is limited to two nights, with regard to the normal sleep pattern found in children with ADHD in our cohort, we can presume a rather strong impact of extrinsic home-based factors that hardly come into play during a mere two nights of a PSG study. Instability of sleep–wake schedule can also be influenced by psychiatric comorbidities which were not excluded in some studies [18,31]; nevertheless, similar findings were also described after excluding psychiatric disorders [51].

Though our results corroborate the preponderant view that ADHD is unrelated to any specific sleep macrostructure changes, total sleep time or sleep timing, the difference between the results of some PSG studies—whether comprising one or two nights—is not quite clear, nor are the main factors underlying possible sleep–wake irregularity in children with ADHD.

Also under discussion is the relationship between ADHD and sleep disturbances, especially SDB and PLMS/PLMD, as these sleep disorders associated with sleep disruption are known to lead to behavioral and cognitive consequences that may mimic ADHD [16]. As for PLMS/PLMD, a possible shared neurobiological mechanism of dopaminergic dysfunction is envisioned [20,52]. In our study we refrained from excluding children with sleep disorders suspected on the basis of subjective data; hence, we believe that despite the relatively small size of our cohort the data on the presence of SDB and PLMS may well reflect their rate of occurrence in ADHD children free from any other comorbidities as well as in the rest of the child population.

The incidence of SDB defined as $AHI > 1$ showed no significant difference between the groups or between the total AI and AHI. Although a correlation of ADHD and SDB was found mainly in questionnaires [23,24,53] and even PSG studies [20,25], most objective studies stop short of corroborating such connection [10,12,21,28,34,54,55]. Some authors [8,22,23,42] reported particularly the milder forms of SDB rather than the more severe ones to be more common in children with ADHD. Since we chose $AHI > 1$ as the cut-off value for abnormality and yet found no difference between children with ADHD and controls, we cannot support this hypothesis. In what is a certain limitation of our effort, neither esophageal pressure measurement nor nasal cannula/pressure transducer was used for SDB rating; consequently, we could not rule out upper airway resistance syndrome.

An association between PLMS/PLMD and ADHD was emphasized by Picchiatti et al. [27], who found a significantly higher incidence of PLMD among children with ADHD [27,28], and corroborated by other authors [20,25,30,31]. To go by the conclusions of a meta-analysis of ADHD and sleep, the only significant difference between children with ADHD and controls is a greater incidence of PLMS/PLMD in ADHD [14]. In our cohort, we found no increased incidence of PLMS ($PLMI > 5$) in children with ADHD compared to the controls, which is in agreement with some other studies [8–10,12,34]. Taking into account that we made use of the newly recommended criteria for PLM duration (0.5 up to 10 s), the rate of PLMS in the controls during the first night was approximately consistent with the highest reported prevalence in a child population (i.e., 12–16.5%) [56,57]. Even though night-to-night variability of PLMS was observed in the groups under study, the differences never reached significance. However, a significant difference

in the inter-night trend of PLMI was apparent (a PLMI decrease in the ADHD group and a PLMI increase in the control group during the second night). PLMS variability across the nights had been described previously in adults [58–60], mainly in patients with restless legs syndrome (RLS), as a sign of presumably unstable central dopaminergic activity [58]. Based on our own observation, we can suggest that night-to-night variability of PLMS may also occur in children with ADHD, indeed, throughout the child population, as only recently described in a study of PLMS in children [61]. Future studies should assess the possible influence of this phenomenon on reported increased incidence of PLMS in children with ADHD.

The hypothesis of insufficient vigilance being that the underlying pathology in ADHD was postulated by Weinberg [62]. In this concept, excessive motor activity is seen as a strategy to stay awake and alert. Hypoarousal could be induced via the same mechanisms as the core ADHD symptoms, i.e., through catecholamine (mainly dopaminergic) dysfunction, functional and structural abnormalities of prefrontal cortex and its networks with other brain regions. Corroboration of this theory can be seen in successful treatment of ADHD with central nervous system stimulants which increase catecholamine activity in the brain and induce alertness. Increased daytime somnolence in ADHD was proved in two studies using MSLT [19,20]. In our study we were not able to replicate these results. The MSLT duration was set at 20 min as it was used in most studies of daytime sleepiness in children [46–49], including a study which proved increased daytime somnolence in patients with ADHD [19]. While we cannot rule out that slight differences in daytime sleepiness may have been missed by using a 20-min instead of a 30-min nap, as recommended by some authors at this age [20,63], there is also another study of ADHD which fell short of proving a longer sleep latency or increased number of children who would fall asleep even during the 30-min MSLT [64]. Another reason our results came out differently may lie in the absence of an adaptation night (with polysomnographic recording) in either of the two studies showing shortened sleep-onset latency in children with ADHD [19,20]. In our study a significant first-night effect was observed in the ADHD group. This fact could possibly influence sleep latencies on MSLT in children with ADHD if measured after a single night of polysomnography. A similar first-night effect was reported by Palm et al. [64], who also reported finding no significant inter-group changes in MSLT results (although their study group was rather small and not defined according to DSM criteria). On the other hand, this protocol of our study may have had a bearing on the relatively shorter sleep latencies in both study groups (the impact of sleep laboratory environment for two nights, deprivation of natural stimuli).

With respect to the MSLT results, a potential limitation of our study could be seen in the ADHD group composition. Using a detailed analysis of MSLT results and the indices of Conners' Parent Rating Scale, Lecendreux et al. [19] reported that children with the hyperactive–impulsive subtype of ADHD were especially prone to falling asleep rapidly. The authors conclude that the number and rapidity of MSLT-measured sleep onset are pertinent physiological indices to discriminate between ADHD subtypes. In our study, the ADHD group had a relatively homogeneous makeup (with a prevalence of the combined subtype). Hence, we cannot rule out that this had an impact on the MSLT results or on the lack of a significant correlation between the MSLT outcome and the scores of psychiatric scales. Future research should be aimed at the potential uses of the MSLT as a diagnostic tool for ADHD patients and for the identification of ADHD subtypes.

Even if we were unable to detect increased daytime sleepiness in children with ADHD, significant variation of daytime vigilance expressed in terms of time-related sleep-latency differences between some MSL tests were observed in this group compared with the controls (significant sleep latency differences between tests 1

and 2, 1 and 4, 1 and 5). Thus, these findings as evidence of variable daytime wakefulness in ADHD with discernible declines in tests 1 and 3 (at 10 a.m. and 14 p.m.) may point to hypothetical arousal dysregulation in children with ADHD.

In compliance with other studies proving the hypoarousal state or dysregulation of arousal [19–21] these findings can substantiate the use of wakefulness-promoting agents in ADHD patients as an alternative therapeutic option. Modafinil, a noradrenergic agent used for the treatment of narcolepsy, was well tolerated and efficient for improvement in the core symptoms of ADHD in some studies [65,66], although its effect during long-term use and in comparison with stimulant treatment has yet to be established [67].

5. Conclusion

In our study, we aimed to exclude potential confounding factors such as psychiatric comorbidities, use of medication or first-night effect. We proved neither changes in the basic sleep parameters nor in the incidence of sleep disorders. Based on these data we believe that even though some children with ADHD may suffer from sleep disorders as another comorbidity, SDB or PLMS/PLMD generally has no substantial or causative role to play in ADHD. Night-to-night variability of the rate of PLMS observed in our cohort should possibly be taken into account in future PLMS assessment of children populations. While we fell short of proving increased daytime sleepiness, we did find evidence of significant daytime vigilance variability in children with ADHD, which is in support of the theory of hypoarousal in ADHD.

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